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The Editor-in-Chief
Information Processing & Management
Elsevier

Re: Submission of Original Research Article
Title: "Epistemic Fortitude: Architectural Specialization for Sycophancy Mitigation in Software Engineering Agents"

Dear Editor-in-Chief,

We are pleased to submit our original research article, **"Epistemic Fortitude: Architectural Specialization for Sycophancy Mitigation in Software Engineering Agents,"** for consideration for publication in *Information Processing & Management*.

Significance and Fit with Journal Scope

Large language models (LLMs) have transformed information retrieval, question answering, and knowledge management—core domains of IPM. However, a critical failure mode threatens their reliability: *sycophancy*, where models abandon factually correct answers when users express disagreement. This behavior undermines the trustworthiness of AI-assisted information systems, particularly in high-stakes domains like software engineering where technical correctness is both verifiable and consequential.

Our work addresses this challenge at the intersection of **information science** (trustworthy AI-mediated knowledge access) and **computing** (architectural solutions for reliable AI systems). We present a novel multi-agent architecture that separates epistemic oversight from conversational fluency, achieving robust sycophancy mitigation across diverse LLM families without expensive retraining.

This research aligns strongly with IPM's mission to publish cutting-edge work "at the intersection of computing and information science concerning theory, methods, or applications." Our contribution spans all three: we formalize Epistemic Fortitude as a theoretical desideratum, develop an architectural method for achieving it, and validate the approach through rigorous empirical evaluation.

Novel Contributions

This manuscript makes the following original contributions to the field:

1. **Architectural Innovation**: A Primary-Arbiter framework that decouples truthfulness enforcement from conversational fluency through specialized agents with distinct epistemic roles.
2. **Cross-Model Generalization**: Evaluation across four model families (Gemini, GPT-5.1, Claude Sonnet 4.5, Llama 3.3 70B) spanning 3,200+ conversations, demonstrating architectural effectiveness independent of model provider.
3. **Methodological Rigor**: A dual-experiment protocol ("Agent-Right" vs. "User-Right" scenarios) proving the architecture maintains epistemic fortitude while preserving flexibility to accept valid corrections—evidence of principled evaluation rather than blind stubbornness.
4. **Ablation Study**: Decomposition of improvement into prompt content versus architectural separation, revealing that for highly sycophantic models, architecture accounts for up to 77% of the gains beyond what instructions alone provide.
5. **Sensitivity Analysis**: Robustness testing across multiple threshold configurations, confirming findings hold across reasonable methodological choices.

All results achieve statistical significance ($p < 0.001$) with effect sizes ranging from Cohen's $d = 0.40$ to $d = 1.79$, representing medium to very large practical improvements.

Relevance to IPM Readership

This work directly benefits the IPM community in several ways:

- **Information Retrieval Researchers**: Understanding how conversational AI systems handle contradictory information is crucial for next-generation search and question-answering systems.
- **Knowledge Management Practitioners**: Organizations deploying AI assistants need architectural patterns that ensure epistemic reliability without sacrificing usability.
- **AI Safety Researchers**: Our framework provides a runtime intervention for alignment issues, complementing training-time approaches like RLHF and Constitutional AI.
- **Software Engineering Tool Developers**: The evaluation on SWE-bench tasks demonstrates practical applicability to code review, debugging, and development assistance workflows.

Declarations

We affirm that this manuscript:

- Is original work that has not been published previously in any peer-reviewed venue
- Is not currently under consideration elsewhere
- Will not be submitted to another journal while under review at IPM
- Represents substantial and novel advancement beyond existing literature
- Has been read and approved by all authors
- Contains no plagiarism or fabricated data
- Adheres to ethical standards for research conduct

All authors have directly participated in the conception, execution, analysis, or writing of this study and have no conflicts of interest to declare. Reproducibility materials, including complete source code, evaluation scripts, and datasets, will be made publicly available upon publication.

We believe this work represents a significant contribution to understanding and mitigating a fundamental challenge in conversational AI systems. The architectural approach we present is theoretically grounded, empirically validated across multiple models, and immediately applicable to production information systems.

Thank you for considering our manuscript for publication in *Information Processing & Management*. We look forward to the review process and the opportunity to contribute to the journal's distinguished publication record.

Respectfully submitted,

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